

Jacob Harris

Senior Data Engineer

SUMMARY

Senior Data Engineer with 12+ years of experience designing and scaling cloud-native data platforms across **AWS**, **Azure**, and **GCP**. Proven expertise in building **real-time** and **batch data pipelines**, **lakehouse architectures**, **CDC frameworks**, and **governed enterprise data platforms** using Databricks, Spark, Kafka, and modern ELT frameworks. Strong track record of architecting high-availability systems, enabling enterprise-wide data governance (Unity Catalog, RBAC, lineage), and delivering measurable cost optimizations exceeding \$60K annually. Experienced in leading large-scale cloud migrations, implementing end-to-end data infrastructure, and building observability frameworks that support AI, analytics, and business intelligence at scale.

EDUCATION

Bachelor Degree of Computer Science at University of California, Berkeley - May 2013

TECHNOLOGY

- **Cloud Platforms & Services:**
AWS (EC2, S3, RDS, Aurora, Glue, Lambda, Redshift, EMR, MSK, CloudWatch, CloudTrail, CloudFormation, IAM, VPC), Azure (Azure Synapse Analytics, Azure Blob Storage), Google Cloud (BigQuery, Dataflow, Dataproc, Cloud Storage, Cloud Composer), Databricks (Delta Live Tables, Unity Catalog), Snowflake
- **Data Engineering & ETL Frameworks:**
Apache Airflow, Apache Beam, AWS Glue, Databricks Delta Live Tables (DLT), Apache Kafka, CDC Pipelines, ETL/ELT Architecture, Data Migration, RESTful API Integration, Third-Party API Integration, SQL-based Transformations
- **Big Data & Distributed Processing:**
Apache Spark, PySpark, Spark SQL, Spark Streaming, Databricks, AWS EMR, Hadoop (HDFS, Hive, HBase), Cassandra
- **Programming & Query Languages:**
Python, SQL (Redshift, BigQuery, Snowflake, SQL Server, Oracle), PySpark, Scala, Shell Scripting
- **Data Warehousing & Modeling:**
Dimensional Modeling (Star & Snowflake Schema), Data Lakehouse Architecture, Hybrid Batch & Streaming Architectures, Data Migration Strategies, Schema Design, Performance Tuning
- **DevOps, Infrastructure & Monitoring:**
Terraform, AWS CloudFormation, Jenkins, GitHub, CI/CD Pipelines, Datadog, AWS CloudWatch, Resource Tagging & Cost Optimization, Cluster Right-Sizing
- **Data Governance & Security:**
Unity Catalog, RBAC, Data Lineage, Metadata Management (Apache Atlas), IAM Policies, PHI Compliance Controls
- **Business Intelligence & Analytics:**
Tableau, Power BI, SSRS, Advanced SQL Analytics, Dashboard Development

PROFESSIONAL EXPERIENCES

Exoft

Senior Data Engineer | Saint Louis, MO

Nov 2020 – Present

- Architected and built scalable, secure, and efficient data pipelines using cloud-based platforms (**AWS**, **Azure**, **GCP**) to enable automated data ingestion, transformation, and storage for large-scale systems.
- Expert in **AWS services (EC2, S3, RDS, Glue, Lambda, Redshift)**, **Azure (Azure Synapse, Blob Storage)**, and **Google Cloud (BigQuery, DataFlow)**, designing end-to-end solutions that ensure high availability and performance.
- Led the design and development of complex **ETL pipelines** using **AWS Glue**, **Apache Airflow**, and **Apache Beam** for cloud-based data processing and migration tasks between on-premise systems and cloud platforms (**Redshift**, **Snowflake**).
- Optimized batch and real-time data workflows using **Apache Airflow**, **AWS Glue**, and **Google Cloud Composer**, ensuring low-latency processing and automated pipeline orchestration.
- Utilized **Python**, **PySpark**, **Scala**, and **Shell scripting** for transforming large datasets, automating cloud-based **ETL workflows**, and optimizing data pipeline performance in **AWS** and **Azure environments**.
- Managed **complex ELT workflows** using Apache Airflow to orchestrate data loading, transformation, and storage processes in a highly available cloud infrastructure, ensuring that ETL/ELT operations met SLAs.
- Designed and implemented **hybrid ELT architectures** using both batch and real-time data processing frameworks, integrating AWS Redshift, Google BigQuery, and Snowflake for scalable data transformation and analytics.
- Implemented best practices for data governance using **AWS IAM roles**, **VPCs**, **CloudTrail** for security compliance, and metadata management tools such as **Apache Atlas**.
- Architected and delivered the **Member360 near-real-time platform** with a dual-latency design (5-second NRT via Kafka and ~30-minute streaming/batch via Delta Live Tables), writing to Databricks (offline analytics layer) and Aurora RDS (online serving layer). Enabled AI-assisted coaching, marketing segmentation, and unified member profiles across 16+ integrated datasets.
- Enabled Unity Catalog GA across the organization **by migrating ~3,000 DLT tables/views and ~4,000 job tables** from Hive Metastore to Unity Catalog; established RBAC policies, lineage tracking, and governance controls. Unlocked Nexus PHI access for global engineering teams and significantly improved cost observability and data compliance posture.

- Established **end-to-end data infrastructure** by building **20+ CDC pipelines and configuring Databricks workspaces**, DLT pipelines, and compute clusters, enabling consistent development workflows, environment isolation, and reduced deployment risk.
- Designed and implemented a **configuration-driven Datadog monitoring framework** with multi-environment (dev/stage/prod) support and automated deployment via Airflow. Implemented SLA latency tracking, CDC freshness validation, critical table health checks, and Google Sheets monitors — enabling self-serve observability for 6+ downstream teams.
- **Drove \$60K+ annual direct infrastructure cost savings** through Databricks resource tagging, cluster right-sizing (i3en → r5), single-node DLT optimization, and decommissioning redundant AWS MSK resources; contributed to \$1.38M in broader team-wide savings initiatives.
- Developed robust data migration strategies for transferring data between **HDFS**, **AWS S3**, and cloud data warehouses like **Snowflake** and **Redshift**, optimizing query performance and reducing data retrieval time.
- Collaborated with cross-functional teams to implement data models in **Redshift** and **Snowflake**, improving schema designs to better align with business needs and analytics requirements.
- Integrated third-party APIs for automated data extraction and validation in **JSON** format using **Python scripts**, ensuring high data integrity.
- Built business intelligence reports and interactive dashboards in **Tableau**, providing actionable insights for senior stakeholders.
- Leveraged **AWS CloudWatch** and **Datadog** for monitoring data pipelines and ensuring robust data quality and real-time monitoring of cloud infrastructure.
- Utilized Terraform and **AWS CloudFormation** to automate cloud resource provisioning and scaling, streamlining deployment cycles.
- Designed and implemented serverless computing solutions using **AWS Lambda** and **Google Cloud Functions**, achieving event-driven data processing automation.

Orases

Data Engineer | New York City, NY

Aug 2017 – Sep 2020

- Played a key role in migrating legacy data systems (**Oracle**, **Teradata**, **Sybase**) to cloud-based data warehouses (**AWS Redshift**, **Google BigQuery**) using complex ETL pipelines and cloud-based data services.
- Led the development of cloud data architecture by optimizing **ETL processes** on **AWS** (**EMR**, **S3**, **Redshift**, **Glue**) and Google Cloud (**Dataproc**, **BigQuery**), improving system scalability and performance.
- Utilized **Apache Spark** (**PySpark**, **Spark SQL**) and **AWS Glue** to develop data transformation solutions for large-scale data processing, supporting business intelligence needs.
- Migrated large datasets from on-premise relational databases (**Oracle**, **SQL Server**) to cloud-based infrastructure, ensuring fast and efficient data processing using **AWS EMR** and **Google Dataproc**.
- Developed real-time data ingestion pipelines integrating third-party APIs using **Python** and **Apache Airflow**, ensuring timely delivery of business-critical data into cloud data warehouses.
- Collaborated with business intelligence teams to develop complex SQL queries for data validation, reporting, and analytics, optimizing performance in cloud environments (**BigQuery**, **Redshift**).
- Built automated CI/CD pipelines using **Jenkins**, **GitHub**, and **Terraform**, improving the efficiency and reliability of deployment processes.
- Utilized data lakes (**HDFS**, **Hive**) and cloud storage solutions (**Google Cloud Storage**, **AWS S3**) for data management, ensuring high availability and fault tolerance.
- Designed and implemented data validation and quality checks using Informatica **Data Quality (IDQ)**, ensuring accurate data pipelines and reporting.
- Developed and maintained automated data pipelines for large-scale data ingestion, transformation, and integration using **AWS Glue**, **Apache Airflow**, and **SQL-based transformations**.
- Enhanced data models in **Redshift** and **Google BigQuery**, applying star and snowflake schema designs for improved **BI reporting**.
- Worked on data governance best practices, ensuring compliance and data security in cloud platforms using IAM roles and monitoring tools (**CloudTrail**, **CloudWatch**).
- Automated workflows using **AWS Lambda** and **Apache Airflow**, reducing operational overhead and improving system responsiveness.

InApp Inc.

Data Analyst / Data Engineer | Palo Alto, CA

Jul 2013 – Jul 2017

- Designed **cloud-based ETL pipelines** using **AWS Redshift**, **S3**, **Lambda**, and **EMR**, enabling large-scale data transformations and improving data accessibility for analytics.
- Led data migration efforts from on-premise **Oracle** and **Greenplum** systems to cloud infrastructure (**AWS Redshift**), improving data processing speed and system scalability.
- Developed and maintained **Spark-SQL** applications on **Databricks** for data transformation and aggregation, supporting data-driven decision-making across departments.
- Automated data workflows using **SSIS** and **SQL Server Agent**, ensuring regular updates to business-critical reports and dashboards.
- Developed interactive dashboards and reports using **Power BI** and **SSRS**, streamlining access to key business metrics for leadership and operations teams.
- Built and optimized SQL queries for data analysis and report generation, ensuring data accuracy and performance tuning for large datasets.
- Contributed to architecture discussions, collaborating with product teams, data scientists, and business analysts to define data requirements and business logic.
- Automated data ingestion processes and batch jobs using **Python** and **shell scripting**, improving the efficiency of data workflows.

- Worked with cloud storage services (**AWS S3**, **AWS Glacier**) and NoSQL databases (**Cassandra**) to manage and back up large datasets, ensuring data availability and disaster recovery readiness.
- Leveraged **Pandas**, **PySpark**, and **SQL** for performing data analysis and building actionable reports that supported business intelligence needs.
- Implemented best practices for data management, ensuring that reporting and analysis systems adhered to company data governance policies.
- Optimized database performance by analyzing SQL execution plans and recommending improvements in query structures.